End of Term Project Documentation – SWAPD 402 (Mobile Application Development)

This document outlines the end-of-term project for the **Mobile Application Development (SWAPD 402)** course. Students are required to develop a complete mobile application using **Flutter and Dart.** The project will be divided into three phases, each reflecting the technical and design skills learned throughout the semester.

Project Idea Selection Criteria:

1. The project must be a mobile application developed using Flutter and Dart.
2. It should solve a real problem or serve a useful purpose for users (e.g., educational, health, entertainment, productivity, or travel).
3. The app should be scalable across the three phases — starting from a basic version and growing into a complete functional product.
4. It should demonstrate a balance between frontend UI/UX and backend logic or data handling.
5. Use of device capabilities (camera, GPS, notifications, etc.) is encouraged.

Phase 1 – App Concept and Basic UI (Week 5)

Students should start by implementing the foundation of their mobile app. This phase focuses on setting up the app project, navigation, and user interface elements.

1. Create the basic Flutter app structure and main navigation flow (e.g., bottom navigation or tab bar).
2. Design and implement at least three functional screens (e.g., home, profile, settings).
3. Use appropriate Flutter widgets for layout, text, buttons, and forms.
4. Ensure responsiveness across different screen sizes.
5. Demonstrate good design practices (color schemes, typography, spacing).
6. Project Idea and Documentation :

    - A clear description of the app idea, including its main purpose, target users, and key use cases.
    - A development **plan outlining milestones**, the app's main modules or screens, and how features will evolve across the phases.
    - This document should be concise (1–2 pages) and demonstrate that the team has a structured plan for completing the app within the semester.

Phase 2 – Core Functionality and Data Handling (Week 8)

This phase focuses on implementing logic, data management, and interactivity within the app using the concepts covered in labs and coursework.

1. Implement data storage using SQLite, SharedPreferences, or Firebase.
2. Apply state management (e.g., Provider or Context API) to handle app logic.
3. Integrate at least one API or network call for fetching or sending data.
4. Add form handling, input validation, and user feedback mechanisms.
5. Include one device feature (e.g., camera, GPS, local notifications, or sensors).

Phase 3 – Finalization, Optimization, and Deployment (Week 12)

In the final phase, students will polish, optimize, and deploy their mobile applications.

1. Add animations and transitions to enhance user experience.
2. Optimize performance and fix bugs using debugging tools.
3. Implement Firebase authentication or cloud synchronization (optional but recommended).
4. Add app icon, splash screen, and prepare the final UI.
5. Generate and submit a release APK/IPA file ready for deployment.
6. Final project documentation :

    - A comprehensive report summarizing the entire project, including :

        - The app's purpose, target users, and overall functionality.
        - All features implemented across the three phases.
        - The tools, technologies, and frameworks used (e.g., Flutter, Firebase, APIs).
        - Screenshots or interface previews of the final app.
        - The document should be well-organized (typically 3–5 pages) and written in a clear, professional format suitable for submission alongside the final project.

General Notes: Teams should consist of 3-4 students. All code must be original and well-documented with inline comments. UI/UX design quality and interactivity will be key grading factors. Each team will present and demo their project in Week 15.